

GRANIL 3mg/3mL Solution for Injection/ Infusion



A1. Name of Product

GRANIL 3mg/3mL Solution for Injection/ Infusion

A2. Name and Strength of Active Ingredient(s)

Granisetron Hydrochloride 3.36 mg/3 mL (eq. to Granisetron free base 3 mg/3 mL)

A3. Product Description

Clear, colorless liquid.

Color of preseutation after reconstitution/ dilution: A colorless clear solution.

A4.1. Pharmacodynamics

Granisetron Hydrochloride is a selective antagonist of 5-hydroxytryptamine (3) (5-HT(3)) receptors present peripherally on vagal nerve terminals and centrally in the chemoreceptor trigger zone. Binding results in blockade of serotonin stimulation and subsequent vomiting triggered by emetogenic stimuli. Granisetron Hydrochloride has little or no affinity for other serotonin receptors, for adrenergic, for dopamine-D (2), histamine-H (1), benzodiazepine, picrotoxin or opioid receptors.

A4.2. Pharmacokinetics

Absorption

Granisetron is rapidly absorbed after oral doses and peak plasma concentrations occur after about 2 hours. Oral bioavailability is about 60% as a result of first-pass hepatic metabolism. Oral bioavailability is generally not influenced by food.

Distribution

Granisetron has a large volume of distribution of around 3 litres/kg

The protein binding of granisetron is 65%

Granisetron distributes freely between plasma and red blood cells

Granisetron is widely distributed in body tissues

Volume of Distribution 0.8 to 10.4 L/kg

Metabolism

Metabolism involves N-demethylation and aromatic ring oxidation followed by conjugation.

Metabolism is most likely mediated by the cytochrome P-450 3A subfamily.

Granisetron appears to be metabolized in the liver.

Approximately 11% of the orally dose is eliminated unchanged in the urine within 48 hours.

The remainder of the dose is excreted as metabolites, 48% in the urine and 38% in the feces

Metabolites may have 5-HT3 receptor antagonist activity

Excretion

Granisetron clearance is not affected by renal impairment, but is lower in the elderly and in patients with hepatic impairment.

Renal Excretion: Approximately 12% of an intravenous dose and 11% of the oral dose is eliminated unchanged in the urine over 48 hours.

Feces: Approximately 83% to 86% of the dose is excreted as metabolites. Of this amount that is excreted as metabolites, 38% of the oral dose and 34% of the I.V. dose are eliminated in the feces.

A5. Indication

Granisetron is indicated for the prevention and treatment (control) of

- a) Acute and delayed nausea and vomiting associated with chemotherapy and radiotherapy
- b) Post-operative nausea and vomiting

A6. Standard Dose

Chemotherapy Induced Nausea and Vomiting (CINV)

Adults

Prevention: A dose of 1-3 mg (10-40 mcg/kg) of Granisetron should be administered either as a slow intravenous injection (over 30 seconds) or as an intravenous infusion diluted in 20 to 50 mL infusion fluid and administered over 5 minutes, prior to the start of chemotherapy.

Treatment: A dose of 1-3 mg (10-40 mcg/kg) Granisetron should be administered either as a slow intravenous injection (over 30 seconds) or as an intravenous infusion diluted in 20 to 50 mL infusion fluid and administered over 5 minutes. Further treatment doses of Granisetron may be administered, if required, at least 10 minutes apart. The maximum dose of Granisetron to be administered over 24 hours should not exceed 9 mg.

Pediatrics

Prevention and Treatment: A dose of 10-40 mcg/kg body weight (up to 3 mg) should be administered as an intravenous infusion, diluted in 10 to 30 ml infusion fluid and administered over 5 minutes prior to the start of chemotherapy. One additional dose may be administered within a 24 hour period if required. This additional dose should not be administered until at least 10 minutes after the initial infusion.

Radiotherapy Induced Nausea and Vomiting (RINV)

Adults

Prevention: A dose of 1-3 mg (10-40 mcg/kg) of Granisetron should be administered either as a slow intravenous injection (over 30 seconds) or as an intravenous infusion diluted in 20 to 50 ml infusion fluid and administered over 5 minutes, prior to the start of radiotherapy.

Treatment: There is insufficient information to recommend the intravenous administration of Granisetron in the treatment of RINV in adult patieuts.

Pediatrics

There is insufficient information to recommend intravenous administration of Granisetron in the prevention and treatment of RINV in children.

Post-operative Nausea and Vomiting (PONV)

Adults

Prevention: A dose of 1 mg (10 mcg/kg) of Granisetron should be administered as a slow intravenous injection (over 30 seconds) prior to induction of anesthesia.

Treatment: A dose of 1 mg (10 mcg/kg) of Granisetron should be administered by slow intravenous injection (over 30 seconds). The maximum dose for patients undergoing anesthesia for surgery is a total dose of 3 mg of Granisetron I.V. in one day.

Pediatrics

There is insufficient information to recommend intravenous administration of Granisetron in the prevention and treatment of postoperative nausea and vomiting in pediatric patients.

Special Dosage Instruction

Geriatrics: No dosage adjustments required.

Renal Impairment: No dosage adjustments required.

Hepatic Impairment: No dosage adjustments required.

A7. Mode of Administration

Intravenous injection

A8. Contraindication

Contraindicated in patients hypersensitive to Granisetron or its excipients.

Incompatibilities:

Admixtures of Granisetron Hydrochloride and dexamethasone sodium phosphate are compatible at concentrations of 10 to 60 mcg/mL granisetron and 80 to 480 mcg/mL dexamethasone phosphate in either 0.9% sodium chloride or 5% glucose intravenous infusion fluids.

A9. Warning and Precautions

As Granisetron may reduce lower bowel motility, patients with signs of sub-acute intestinal obstruction should be monitored closely following administration of Granisetron.

As for other 5-HT3 antagonists, cases of ECG modifications including QT prolongation have been reported with Granisetron. These ECG changes with Granisetron were minor and generally not of clinical significance, specifically with no evidence of proarrhythmia. However, in patients with pre-existing arrhythmias or cardiac conduction disorders, this might lead to clinical consequences. Therefore, caution should be exercised in patients with cardiac co-morbidities, on cardio-toxic chemotherapy and/or with concomitant electrolyte abnormalities.

A10. Interaction with Other Medicaments

The metabolism of Granisetron is induced by phenobarbital.

Granisetron has been safely administered in humans with benzodiazepines, neuroleptics and anti-ulcer medications, commonly prescribed with antiemetic treatments.

Granisetron has shown no apparent drug interaction with emetogenic cancer chemotherapies.

Granisetron has been safely administered with commonly used anesthetic and analgesic agents.

In patients concurrently treated with drugs known to prolong QT interval and/or are arrhythmogenic, this may lead to clinical consequences.

A11. Pregnancy and Lactation

Pregnancy

No human studies have been conducted on the use of Granisetron during pregnancy. Due to the lack of human reproductive studies, Granisetron should be used during pregnancy only if clearly needed.

Lactation

Infant risk cannot be ruled out. Available evidence and/or expert consensus are inconclusive or is inadequate for determining infant risk when used during breastfeeding. Weigh the potential benefits of drug treatment against potential risks before prescribing this drug during breastfeeding.

A12. Side Effects

Neurologic: asthenia , headache ,somnolence

Cardiovascular: prolonged QT interval

Immunologic: hypersensitivity reaction (rashes and anaphylaxis)

Other: fever, constipation, elevations in hepatic transaminases

Effects on ability to drive and use machines

There has been no evidence from human studies that granisetron has any adverse effect on alertness.

A13. Symptoms and Treatment of Overdose

There is no specific antidote for Granisetron. In the case of over dosage with Granisetron, symptomatic treatment should be given.

Overdosage of up to 38.5 mg of Granisetron Hydrochloride as a single injection has been reported without symptoms or only the occurrence of a slight headache.

A14. Storage Condition

Store at temperature below 30°C, protected from light. Do not freeze. Store at 25°C after dilution and discard within 24 hours.

A15. Shelf Life

2 years

A16. Dosage Form and Packaging Available

3 mL x 5 ampoules



Manufacturer:
YUNG SHIN PHARMACEUTICAL IND. CO., LTD.
Taichung Youth Factory
No. 27, Kung Chiu Rd., Tachia, Taichung, Taiwan, R.O.C.
Importer and Product Registration Holder:
Y.S.P. INDUSTRIES (M) SDN. BHD.
Lot 3, 5 & 7, Jln. P/7, Sec. 13, Kaw. Perindustrian
Bdr. Baru Bangi, 43000 Kajang, Selangor D.E., Malaysia